

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5.				Indicative content: Conduction: Gas (accept vacuum) in the space between the two panes is a poor conductor which reduces conduction through the window. Laying fibreglass insulation on the attic floor has pockets of air trapped in it that reduce heat loss by conduction (air is a good insulator). Convection: Fibreglass insulation reduces the heat energy getting into the loft which reduces convection currents in there. Radiation: The reflective coating on the glass reduces heat energy loss by radiation either by reflection or emission. The surfaces of the insulation are shiny to reflect radiant heat back into the house and to reduce heat radiated into the loft. 5–6 marks Comprehensively describes the reduction in heat loss by all heat transfer methods for either improvement. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3–4 marks Comprehensively describes the reduction in heat loss by two of the heat transfer methods for either improvement or gives a limited description of all heat transfer methods. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i>	6			6		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				1–2 marks Gives a limited description of the reduction in heat loss by one or two of the heat transfer methods for either improvement. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
				Question 5 total	6	0	0	6	0	0